Scalar k:2k degenerate-face theorem, projective boundary, and mixed

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Purpose and scope

The bare stationary scalar $k{:}2k$ square-first inequality is now proved on both degenerate frequency faces and reduced, on the genuinely mixed face, to an explicit compact-interior problem with certified tails. The mixed all-q theorem and every production conclusion remain open.

R-110 computed the exact complete two-frequency packet P, its whole-output realized covariance square H, and the strict right-small-q margin. This note first takes the exact scaling quotient. On $w = 0$, a linear exponential tilt followed by a one-sided Bennett/floor split proves $\log \mathbb{E}e^{-qP} \leq q^2 H/4$ for every $q \geq 0$ and every amplitude. The $v = 0$ face is the same theorem after rescaling. This is a genuine all-q result, not a finite quadrature statement.

For $v, w > 0$, the phase is integrated exactly into I_0. The dangerous $A \to \infty$, $q \to 0$ projective corner is safe: its limiting gap is the sum of two positive centered-exponential gaps, and the complete first inverse- amplitude correction has a coefficientwise positive numerator. The exact phase minimum gives a finite algebraic candidate set and removes all sufficiently large q. A square decomposition supplies explicit factorized tails, while a sharp Bessel comparison supplies a second globally integrable majorant.

Three tempting shortcuts are retired. The quadratic Bessel envelope is nonintegrable, conditioning one frequency produces negative effective scalar coefficients for which positive-coefficient tensorization is false, and the tilted variance need not decrease. None of these failures is a counterexample to the mixed all-q target.

R-103 already closes REG in its declared regular scope, and R-087 already closes the fixed-cutoff variational CORE. The old R-085 equations (4.10)–(4.11) and (6.5) are not reopened or retroactively proved here. The live production theorem is still a subdivision-invariant full-A1 conditional cluster estimate plus a one-use source/sextic aggregation ledger. Hence $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, removals, an interacting measure, and Sector-A closure remain open at T4.

The scalar model is the R-110 stationary circle model. It is not asserted to be an embedded full A1 six-real production cluster. Polynomial identities are exact. The two numerical fixtures are adversarial verification data only: one demonstrates near-tightness on an exactly evaluable boundary, and one falsifies a proof route while remaining far inside the target. Neither is used to certify the unbounded mixed parameter domain.

1 Exact scaling quotient and radial normal form

Let

$$X(x) = A + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x, \tag{1.1}$$

where $(a_1, b_1) \sim N(0, v\mathrm{I}_2)$ and $(a_2, b_2) \sim N(0, w\mathrm{I}_2)$ independently. As in R-110, put

$$R = \frac{a_1^2 + b_1^2}{4}, \qquad S = \frac{a_2^2 + b_2^2}{4}, \qquad C = \mathrm{Re}(z^2 \bar{u}), \qquad \gamma = v + 4w. \tag{1.2}$$

The complete covariance-normal packet is

$$\begin{aligned} P = A^2 R + 4A^2 S - \frac{\gamma A^2}{2} + 6AC \\ + R^2 + 10RS + 4S^2 - \gamma(R + S), \end{aligned} \tag{1.3}$$

and

$$\begin{aligned} H = A^4 \left(\frac{v^2}{2} + 8w^2 \right) + A^2 (v^3 + 10v^2 w + 16vw^2 + 16w^3) \\ + \frac{5}{4} v^4 + v^3 w + 25v^2 w^2 + 16vw^3 + 20w^4. \end{aligned} \tag{1.4}$$

Under

$$A = \sqrt{\lambda}\, \bar{A}, \quad v = \lambda \bar{v}, \quad w = \lambda \bar{w}, \quad (R, S, C) = \left(\lambda \bar{R}, \lambda \bar{S}, \lambda^{3/2} \bar{C} \right), \tag{1.5}$$

one has

$$P \overset{d}{=} \lambda^2 \bar{P}, \qquad H = \lambda^4 \bar{H}. \tag{1.6}$$

Thus only $q\lambda^2$ remains.

Assume first $v > 0$ and define

$$a = \frac{A^2}{v}, \qquad r = \frac{w}{v}, \qquad t = qv^2, \qquad T = \frac{2R}{v}, \qquad U = \frac{2S}{w}. \tag{1.7}$$

For $w > 0$, T, U are independent unit exponentials and the relative phase ϕ is uniform. Then $P = v^2 p$ and $H = v^4 h$, where

$$\boxed{\begin{aligned} p = \frac{T^2}{4} + r^2 U^2 + \frac{5}{2} rTU + a \left(\frac{T}{2} + 2rU - \frac{1 + 4r}{2} \right) \\ - \frac{1 + 4r}{2} (T + rU) + 3\sqrt{ar/2}\, T \sqrt{U} \cos \phi, \end{aligned}} \tag{1.8}$$

$$\boxed{\begin{aligned} h = a^2 \left(\frac{1}{2} + 8r^2 \right) + a(1 + 10r + 16r^2 + 16r^3) \\ + \frac{5}{4} + r + 25r^2 + 16r^3 + 20r^4. \end{aligned}} \tag{1.9}$$

Writing p_0 for (1.8) without its cosine term gives the exact integral

$$\boxed{M(a, r, t) = \mathbb{E} e^{-tp} = \int_0^\infty \int_0^\infty e^{-T - U - t p_0(T, U)} I_0 \left(3t \sqrt{ar/2}\, T \sqrt{U} \right) dT\, dU.} \tag{1.10}$$

This is the canonical mixed compact-core object.

The R-110 local identity becomes

$$h - 2\,\mathrm{Var}(p) = \frac{1 + 16r^2 + 16r^4}{4} > 0. \tag{1.11}$$

Moreover, every coefficient of $\mathbb{E}p^3$ is positive. Therefore

$$\frac{t^2 h}{4} - \log M(a, r, t) = \frac{t^2}{16} (1 + 16r^2 + 16r^4) + \frac{t^3}{6} \mathbb{E}p^3 + O(t^4) \tag{1.12}$$

is positive on a right neighborhood of zero for each fixed shape. Equation (1.12) is not by itself uniform as $a \to \infty$.

2 All-q theorem on both degenerate frequency faces

Theorem 2.1 (exact all-q closure for $vw = 0$)

For every $A \in \mathbb{R}$, $v, w \geq 0$ with $vw = 0$, and every $q \geq 0$,

$$\boxed{\log \mathbb{E} e^{-qP} \leq \frac{q^2}{4} H.} \tag{2.1}$$

Proof on $w = 0$. Assume $v > 0$, let

$$\alpha = \frac{A^2}{v}, \qquad X = \frac{2R}{v} \sim \mathrm{Exp}(1), \qquad s = \frac{qv^2}{4}. \tag{2.2}$$

Then

$$P = \frac{v^2}{4} Z_\alpha, \qquad Z_\alpha = 2\alpha(X-1) + W, \qquad W = X^2 - 2X \geq -1, \tag{2.3}$$

and (2.1) is exactly

$$\log \mathbb{E} e^{-sZ_\alpha} \leq (2\alpha^2 + 4\alpha + 5)s^2. \tag{2.4}$$

Tilt first by the centered linear exponential. With

$$\ell = 1 + 2\alpha s, \tag{2.5}$$

one has the exact identity

$$\log \mathbb{E} e^{-sZ_\alpha} = 2\alpha s - \log \ell + \log \mathbb{E}_{\mathrm{Exp}(\ell)} e^{-sW}. \tag{2.6}$$

The first two terms obey

$$2\alpha s - \log(1 + 2\alpha s) \leq 2\alpha^2 s^2. \tag{2.7}$$

Under $\mathrm{Exp}(\ell)$, direct moments give

$$\mathbb{E}_\ell W = -\frac{2(\ell - 1)}{\ell^2}, \qquad \mathrm{Var}_\ell(W) = \frac{4(\ell^2 - 4\ell + 5)}{\ell^4} \leq 8. \tag{2.8}$$

The last inequality follows from

$$8\ell^4 - 4(\ell^2 - 4\ell + 5) = 4(\ell - 1)(2\ell^3 + 2\ell^2 + \ell + 5) \geq 0. \tag{2.9}$$

If $s \geq (4\alpha + 5)^{-1}$, the floor $W \geq -1$ gives

$$\log \mathbb{E}_\ell e^{-sW} \leq s \leq (4\alpha + 5)s^2. \tag{2.10}$$

Together with (2.7), this proves (2.4) on the large-s branch.

If $s \leq (4\alpha + 5)^{-1} \leq 1/5$, put $Y = \mathbb{E}_\ell W - W$. Then $\mathbb{E}_\ell Y = 0$, $Y \leq 1$, and $\mathbb{E}_\ell Y^2 \leq 8$. The one-sided exponential estimate gives

$$\begin{aligned} \log \mathbb{E}_\ell e^{-sW} &= -s\mathbb{E}_\ell W + \log \mathbb{E}_\ell e^{sY} \\ &\leq \frac{4\alpha s^2}{\ell^2} + 8(e^s - 1 - s) \\ &\leq \left(4\alpha + \frac{30}{7}\right) s^2 < (4\alpha + 5)s^2. \end{aligned} \tag{2.11}$$

Here the last estimate uses $e^s - 1 - s \leq s^2/[2(1 - s/3)]$ for $0 \leq s \leq 1/5$. Equations (2.7) and (2.11) prove (2.4) on the small-s branch.

The $v = 0$ face. If $w > 0$, set $X = 2S/w$ and $\alpha = A^2/w$. Then

$$P = w^2 Z_\alpha, \qquad H = w^4(8\alpha^2 + 16\alpha + 20). \tag{2.12}$$

With $s = qw^2$, inequality (2.1) is again exactly (2.4). Finally, $v = w = 0$ is trivial. $\square$

Boundary closed form. On $w = 0$, direct integration also gives, with $t = qv^2$ and $\beta = 1 + t(\alpha - 1)/2$,

$$\boxed{\log M_0(\alpha, t) = \frac{t\alpha}{2} + \frac{1}{2}\log\frac{\pi}{t} + \frac{\beta^2}{t} + \log \operatorname{erfc}\left(\frac{\beta}{\sqrt{t}}\right).} \tag{2.13}$$

The exact rational fixture $\alpha = 27$, $t = 1/9000$ has

$$\frac{t^2 h}{4} - \log M_0 = 2.16867529503112319514537025 67\ldots \times 10^{-9} > 0, \tag{2.14}$$

and ratio $0.9982109464147929\ldots$. This demonstrates the near-tight large-amplitude/small-q corner; it is not needed in the proof of Theorem 2.1.

3 Large-amplitude projective boundary

Write

$$p = aL + \sqrt{a}\, B + Q, \tag{3.1}$$

where

$$L = \frac{T-1}{2} + 2r(U-1), \qquad B = 3T\sqrt{rU/2}\cos\phi, \tag{3.2}$$

$$Q = \frac{T^2}{4} + r^2U^2 + \frac{5}{2}rTU - \frac{1+4r}{2}(T + rU). \tag{3.3}$$

Theorem 3.1 (positive projective limit)

Fix $x, r \geq 0$ and set $t = x/a$. Then

$$M(a, r, x/a) \longrightarrow M_\infty(x, r) = \frac{e^{x(1+4r)/2}}{(1 + x/2)(1 + 2rx)}. \tag{3.4}$$

The limiting target gap is

$$\boxed{D_0(x, r) = d(x/2) + d(2rx), \qquad d(y) = \frac{y^2}{2} - y + \log(1 + y) \geq 0.} \tag{3.5}$$

Indeed,

$$d'(y) = \frac{y^2}{1 + y} \geq 0, \qquad d(0) = 0. \tag{3.6}$$

Thus $D_0(x, r) > 0$ for $x > 0$. □

Proposition 3.2 (positive first inverse-amplitude correction)

For fixed x, r,

$$\frac{x^2 h}{4a^2} - \log M(a, r, x/a) = D_0(x, r) + \frac{D_1(x, r)}{a} + O(a^{-2}), \tag{3.7}$$

where

$$\boxed{D_1(x, r) = \frac{x^3 \Pi(x, r)}{4(x + 2)^2(1 + 2rx)^2} > 0} \tag{3.8}$$

for $x > 0$, and

$$\begin{aligned} \Pi = {} & 64r^5x^3 + 256r^5x^2 + 256r^5x + 64r^4x^3 + 320r^4x^2 + 512r^4x + 256r^4 \\ & + 40r^3x^3 + 224r^3x^2 + 352r^3x + 128r^3 + 4r^2x^3 + 56r^2x^2 + 172r^2x + 112r^2 \\ & + 4rx^2 + 26rx + 38r + x + 4. \end{aligned} \tag{3.9}$$

Proof. Under the limiting tilted product law, T and U remain independent exponentials with rates $1+x/2$ and $1+2rx$. Since the phase mean of B vanishes,

$$\log M = \log M_\infty + \frac{1}{a}\mathbb{E}_x\left[\frac{x^2}{2}B^2 - xQ\right] + O(a^{-2}). \tag{3.10}$$

Here the remainder is analytic rather than formal. In the unscaled $v=1$ radial variables, for $a \ge 20\gamma$,

$$a + 10S - 6\sqrt{aS} \ge \frac{a}{10}, \qquad p_- \ge R^2 + \frac{a}{20}R + 4S^2 + \frac{79a}{20}S - \frac{\gamma a}{2}. \tag{3.10a}$$

At $t = x/a$ this lower bound gives an a-uniform exponential dominator for the integrand and for every polynomial factor used below. With $\varepsilon = a^{-1/2}$ the phase-averaged integrand is even in ε (shift ϕ by π), so dominated Taylor expansion through ε^2 has remainder $O(\varepsilon^4) = O(a^{-2})$ for fixed x, r. Insert the exponential moments $\mathbb{E}T^n = n!/(1+x/2)^n$ and $\mathbb{E}U^n = n!/(1+2rx)^n$, subtract the linear coefficient of h, and clear the positive denominators. The result is (3.8)–(3.9), whose coefficients are all positive. ☐

The leading floor-to-$\sqrt{h}$ shape satisfies

$$2 + 32r^2 - (1+4r)^2 = (4r-1)^2 \ge 0. \tag{3.11}$$

The unique saturation ratio is $r = 1/4$, but then

$$M_\infty(x, 1/4) = \frac{e^x}{(1+x/2)^2}, \qquad D_0(x, 1/4) = 2d(x/2) > 0. \tag{3.12}$$

Proposition 3.2 is a pointwise projective expansion. A uniform remainder bound is still required before one may replace the entire large-a region by a finite certified threshold. This boundary is explicit in the next-action list and prevents an overclaim of compactness already completed.

4 Exact phase floor, large-q cutoff, and rigorous tails

Use the normalized $v=1$ variables $R = T/2$, $S = rU/2$ and put $y = \sqrt{S}$. Minimizing the phase gives

$$\begin{aligned} p_-(R,S) &= R^2 + 10RS + 4S^2 + (a-\gamma)R + (4a-\gamma)S \\ &\quad - 6\sqrt{a}\,R\sqrt{S} - \frac{\gamma a}{2}, \qquad \gamma = 1+4r. \end{aligned} \tag{4.1}$$

Theorem 4.1 (finite algebraic floor and exact high-q cutoff)

For fixed y, the minimizing radial coordinate is

$$R_*(y) = \frac{1}{2}[\gamma - a - 10y^2 + 6\sqrt{a}\,y]_+. \tag{4.2}$$

Consequently

$$\boxed{p_* = \min_{y\ge 0}\left[4y^4 + (4a-\gamma)y^2 - \frac{\gamma a}{2} - \frac{1}{4}[\gamma - a - 10y^2 + 6\sqrt{a}\,y]_+^2\right]}. \tag{4.3}$$

The candidate set consists of $y=0$, the roots of the positive-part bracket, the inactive quartic vertex, and the nonnegative roots of

$$\boxed{84y^3 - 90\sqrt{a}\,y^2 + (20a - 8\gamma)y + 3\sqrt{a}(\gamma - a) = 0.} \tag{4.4}$$

Since $p \ge p_*$,

$$\log M(a,r,t) \le -tp_*.$$

Therefore

$$\boxed{t \ge \frac{4(-p_*)}{h} \quad \Longrightarrow \quad \log M(a,r,t) \le \frac{t^2 h}{4}.} \tag{4.5}$$

Thus no fixed shape can fail at arbitrarily large t. ☐

Theorem 4.2 (two globally integrable majorants)

First, the modified Bessel function obeys

$$\boxed{\log I_0(z) \le \sqrt{4+z^2} - 2, \qquad z \ge 0.} \tag{4.6}$$

To prove it, put $g = I_1/I_0$ and $h = z/\sqrt{z^2+4}$. The Riccati identity is $g' = 1 - g/z - g^2$. At the origin,

$$g = \frac{z}{2} - \frac{z^3}{16} + \frac{z^5}{96} + O(z^7), \qquad h = \frac{z}{2} - \frac{z^3}{16} + \frac{3z^5}{256} + O(z^7),$$

so $h - g = z^5/768 + O(z^7) > 0$ initially. At a first positive contact,

$$g' - h' = \frac{4\sqrt{z^2+4} - z^2 - 8}{(z^2+4)^{3/2}} < 0, \tag{4.7}$$

because

$$(z^2+8)^2 - 16(z^2+4) = z^4 > 0. \tag{4.8}$$

An upward first crossing is impossible, so $g \le h$. Integrate from zero to obtain (4.6). This bound has the correct quadratic/quartic expansion at zero and linear growth at infinity.

Second, let $b = 3\sqrt{ar/2}\,T\sqrt{U}$ and let $p_- = p_0 - b$. Exact square completion gives

$$\boxed{p_- \ge \frac{T^2}{16} + \frac{r^2U^2}{2} - C(a,r),} \tag{4.9}$$

where

$$C(a,r) = \frac{a\gamma}{2} + (\gamma - a)_+^2 + \frac{1}{2}\left(7a + \frac{\gamma}{2}\right)^2. \tag{4.10}$$

Indeed, after subtracting the right side of (4.9), the low-a branch is the sum

$$\left(\frac{T}{2\sqrt{2}} - 3\sqrt{arU}\right)^2 + \left(\frac{T}{4} - (\gamma - a)\right)^2 + \frac{1}{2}\left(rU - 7a - \frac{\gamma}{2}\right)^2 + \frac{5}{2}rTU, \tag{4.11}$$

and on $a \ge \gamma$ the middle square is replaced by $T^2/16 + (a-\gamma)T/2$.

Since $I_0(tb) \le e^{tb}$, the exact integrand in (1.10) is at most

$$e^{tC(a,r)} e^{-T - tT^2/16} e^{-U - tr^2U^2/2}. \tag{4.12}$$

Every rectangular tail is therefore bounded by products of explicit

$$\int_L^\infty e^{-z-cz^2}\,dz = \frac{\sqrt{\pi}}{2\sqrt{c}} e^{1/(4c)} \operatorname{erfc}\left(\sqrt{c}L + \frac{1}{2\sqrt{c}}\right). \tag{4.13}$$

Equations (4.3)–(4.13) provide the algebraic high-t and radial-tail pieces of a finite-box certificate. Completion still requires a uniform projective remainder together with uniform local-t and two-chart face patching as $r \to 0$ and $r \to \infty$. $\square$

5 Failed routes and their exact boundaries

Proposition 5.1 (separated conditional domination fails)

Three direct separations cannot prove the mixed theorem.

Quadratic Bessel envelope. The valid inequality $I_0(z) \le e^{z^2/4}$ inserts

$$+9q^2A^2R^2S \tag{5.1}$$

into the exponent. Along $R = S = L$, this is positive cubic growth, while the original coercive part is only $-15qL^2$. The proposed upper integral is infinite whenever $qA \neq 0$. The sharp linear-growth bound (4.6) is required.

Dropping the S quartic. The elementary transform $\int_0^\infty e^{-\alpha S} I_0(c\sqrt{S})dS$ requires

$$\alpha = \frac{2}{w} + q(4A^2 - \gamma + 10R) > 0. \tag{5.2}$$

At $v = w = q = 1$, $A = R = 0$, it equals -3. Thus one cannot discard $4qS^2$ before applying that transform.

Conditional scalar tensorization. Conditioning on S creates the effective scalar coefficient

$$\alpha_{\text{eff}} = \frac{A^2 + 10S - 4w}{v}, \tag{5.3}$$

which is unbounded below across mixed shapes. For $\alpha_{\text{eff}} = -B$,

$$Z_{-B} = (X - B - 1)^2 - (B^2 + 1). \tag{5.4}$$

For fixed $0 < s < 1/2$, exact completion of the one-dimensional integral gives

$$\log \mathbb{E} e^{-sZ_{-B}} = sB^2 - B + O(1), \tag{5.5}$$

whereas blindly extending the positive-α proxy would give

$$(2B^2 - 4B + 5)s^2 = 2s^2B^2 + O(B). \tag{5.6}$$

Because $s - 2s^2 > 0$, (5.6) is false for large B. Theorem 2.1 may not be applied after this conditioning. □

The first and third failures are registered as

$$\begin{gathered}\texttt{NG-2026-07-28-A13-K2K-QUADRATIC-BESSEL-UPPER-DOMINATION},\\ \texttt{NG-2026-07-28-A13-K2K-CONDITIONAL-SCALAR-TENSORIZATION}.\end{gathered} \tag{5.7}$$

Proposition 5.2 (tilted variance need not decrease)

Let $\psi(t) = \log \mathbb{E} e^{-tp}$. Then

$$\psi''(t) = \operatorname{Var}_t(p), \qquad \psi'''(t) = -\mathbb{E}_t[(p - \mathbb{E}_t p)^3]. \tag{5.8}$$

A tempting proof would show $\psi''(t) \le \psi''(0)$ for all $t \ge 0$. It is false. At the exact declared fixture

$$a = 0, \qquad r = 7, \qquad t = \frac{1}{10}, \tag{5.9}$$

adaptive integration after exact elimination of T gives

$$\begin{aligned} \mathbb{E}_t p &= -53.84754430669265763309\ldots,\\ \operatorname{Var}_t(p) &= 440.65392463518411475448\ldots,\\ \mathbb{E}_t[(p - \mathbb{E}_t p)^3] &= -24382.80109039520660173\ldots. \end{aligned} \tag{5.10}$$

Independent tensor Gauss–Laguerre orders 96 and 128 retain the same negative sign. Hence $\psi'''(t) > 0$ there. The target gap at this fixture is $132.60095258\ldots > 0$, so this is a proof-route no-go, not a counterexample.

This failure is registered as

$$\texttt{NG-2026-07-28-A13-K2K-TILTED-VARIANCE-MONOTONICITY}. \tag{5.11}$$

6 Correct owner map: REG, OVERLAP, and CORE

Theorem 6.1 (what the scalar theorem can and cannot own)

Theorem 2.1 closes exactly the $vw = 0$ faces of the R-110 scalar target. A future proof of the mixed face, if uniform in strict-past A, v, w, would give a legal one-fresh-root scalar cluster normalizer after the square is conditionally averaged as required by R-109. It would not by itself prove a production owner theorem.

REG is already closed in its scope. R-103 proves the complete regular H_N and REG lower forms using the exact nonduplicating owners from R-080–R-102. The old R-085 averaged Cartan atom (4.11) and old rational form (6.5) are explicitly non-load-bearing there. Adding the scalar packet as a ninth REG owner would risk duplication and is forbidden.

The old R-085 statements are not consequences. R-085 (4.10)–(4.11) is an OU/output-shell decay contract with Π_m, $m-k$ decay, and a q_k ledger. The scalar theorem has none of these objects. R-108 also corrects the live sufficient Cartan bridge to an unweighted positive-gain statement. The old rational (6.5) was replaced by the exact R-101 endpoint

$$\Delta W_R = R_Q + M_U + K_R. \tag{6.1}$$

Its summands change under representation-preserving subdivision while the complete endpoint does not. The scalar theorem proves no rational owner identity.

CORE is already closed at fixed cutoff. R-087 proves equality of the Boue–Dupuis infimum on the full finite-energy class and the bounded cylindrical-simple core in the declared graph topology. The missing estimate is not another CORE density theorem.

$\mathrm{OVERLAP}_{\mathrm{src}}$ **remains open.** The live target is a cutoff-, chart-, control-, partition-, revisit-, and subdivision-uniform complete source-action lower bound. Its one-root packet must retain complete heat, mean, covariance trace, mixed baseline, rational recovery, every visit, the random-W double divergence, and the full R-063 forest once. The smallest honest local successor has the schematic form

$$\begin{aligned} \mathcal{L}_{j,\mathcal{C}} &:= \log \mathbb{E}_{j-1} \exp[-q(P^{\mathrm{A1}}_{j,\mathcal{C}} + \eta_{j,\mathcal{C}} E_j \\ &\qquad\qquad\qquad\qquad + \zeta_{j,\mathcal{C}} Y_{\mathrm{parent}})], \\ \mathcal{L}_{j,\mathcal{C}} &\le \frac{q^2}{4} \mathbb{E}_{j-1} \|S^{\mathrm{real}}_{j,\mathcal{C}}\|^2_{\mathfrak{S}_2} + r^{\mathrm{pred}}_{j,\mathcal{C}}, \end{aligned} \tag{6.2}$$

with an $\mathcal{F}_{j-1}$-measurable right side and exact subdivision invariance. A separate one-use theorem must then prove

$$\sum_{j,\mathcal{C}} \left\{ \frac{q^2}{4} \mathbb{E}_{j-1} \|S^{\mathrm{real}}_{j,\mathcal{C}}\|^2_{\mathfrak{S}_2} + r^{\mathrm{pred}}_{j,\mathcal{C}} \right\} \le C + \epsilon \sum_j \|h_j\|_2^2 + \delta \|Z_K\|_6^6. \tag{6.3}$$

Only (6.2) plus (6.3), with the accepted owner arithmetic, can feed the R-087 CORE and yield $q = 10/9$ Nelson.

7 Proof and failure map

Route	Verdict	Reusable boundary
Scale quotient and Bessel normal form	proved	Three shape variables (a, r, t) remain in the v chart.
Degenerate frequency faces	proved all-q	Linear exponential tilt plus small-t Bennett and large-t floor.
Right-small-q mixed face	proved	Strict R-110 variance and cubic margins.
Large-amplitude projective face	advanced	Positive limiting gap and positive first inverse-amplitude correction; uniform remainder open.
Large-q face	proved shape-wise	Exact algebraic floor cutoff.
Tail certification	proved	Sharp Bessel comparison and factorized erfc tails.
Quadratic Bessel domination	failed	Positive cubic radial exponent makes the upper integral infinite.
Conditional scalar tensorization	failed	Negative effective coefficient has the wrong B^2 Laplace scale.
Monotone tilted variance	failed	Exact declared fixture has negative tilted third centered moment.
Mixed compact interior	open	Uniform projective remainder and interval or new analytic comparison required.
Full A1 production lift	open	Mean, heat, rational, visits, random-W forest, source, and sextic owners must remain once.

8 Devil's-advocate audit

1. **Objection: the face theorem silently assumes a positive effective coefficient.**

   DISMISSED ON THE DECLARED FACES; UPHELD AGAINST CONDITIONING. Theorem 2.1 has $\alpha = A^2/v \geq 0$ or $A^2/w \geq 0$. Proposition 5.1 proves that conditioning the mixed packet can create $\alpha_{\text{eff}} < 0$, and the face theorem is not applied there.

2. **Objection: the small-t proof reuses the floor budget.**

   DISMISSED. The branches are disjoint. For $s \leq (4\alpha+5)^{-1}$, the centered Bennett estimate spends variance at most eight and $-s\mathbb{E}_\ell W$. For the complementary branch, only $W \geq -1$ is used. Both meet at the declared threshold.

3. **Objection: the projective calculation proves the whole large-amplitude region.**

   VALID WITH AN OPEN MITIGATION. The limiting and first-correction coefficients are exact and positive for fixed (x, r). A uniform remainder over a compactified covariance-shape chart is not yet proved. The mixed all-q theorem remains open.

4. **Objection: the Bessel upper bound has the wrong asymptotic.**

   DISMISSED. Equation (4.6) agrees through the quadratic and quartic terms at zero and grows linearly. The primary and independent certificates check the Riccati first-contact identity and exact comparison square. The quadratic exponential bound is separately rejected as nonintegrable.

5. **Objection: finite quadrature is being used to certify the mixed interior.**

DISMISSED. No finite scan enters a positive theorem. The Gauss–Laguerre calculation only independently confirms the sign of the tilted-variance method fixture, whose primary value comes from a different adaptive one-dimensional integral after exact elimination of T.

6. **Objection: the scalar result closes REG or CORE.**

   DISMISSED AS A SCOPE ERROR. Those results already have their accepted scoped authorities. The scalar theorem is neither an OU Cartan atom nor a rational owner. It is a local cross-resonance normalizer candidate for the still-open full-A1 $\mathrm{OVERLAP}_{\mathrm{src}}$ architecture.

7. **Objection: signs, factors, units, or hard-coded numbers could mask the result.**

   DISMISSED FOR THE CLAIMED SCOPE. Scaling weights make every term in P degree two and every term in H degree four. Both scripts independently reconstruct the normalized packet, projective moments, floor cubic, and tail squares. The primary tilted fixture uses adaptive high-precision integration; the independent route uses tensor Gauss–Laguerre orders 96 and 128. The only decimal assertions are declared adversarial fixtures, not derived theorem constants.

9 Authorities and scope firewall

The load-bearing authorities are R-063 (complete coefficient-jet/Wick forest), R-087 (fixed-cutoff variational CORE), R-101 (nonduplicating rational owner), R-103 (complete regular H_N/REG closure), R-104 (lossless fixed-chart owner assembly and Douglas slack), R-105 (common-root quotient and physical cross-mode boundary), R-108 (complete-cluster quotient/covariance order), R-109 (legal square-first filtration and one-pair theorem), and R-110 (exact random-W and stationary scalar $k{:}2k$ packet).

Proved here are Theorem 2.1, Theorem 3.1, Proposition 3.2, Theorems 4.1–4.2, and the exact owner-map statement of Theorem 6.1. Propositions 5.1–5.2 retire only their named proof architectures. Not proved are the genuinely mixed $v, w > 0$ all-q inequality, a uniform projective remainder, the compact interior interval certificate, a full A1 embedding, a subdivision-invariant production cluster estimate, the one-use aggregation (6.3), $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, cutoff/floor removal, an interacting measure, Sector-A closure, or any tier promotion.

10 Executable verification

The primary certificate uses exact symbolic scaling, exponential moments, projective expansion, floor differentiation, Bessel comparison, tail square completion, the closed-form boundary MGF, and adaptive high-precision tilted moments. The independent certificate imports no primary code. It reconstructs the projective correction separately, verifies the boundary erfc formula against direct adaptive quadrature, and uses tensor Gauss–Laguerre orders 96 and 128 for the tilted method fixture.

The primary contract is 46/46 and the non-importing independent contract is 35/35. The manifest-pinned verifier reruns both children, checks all hashes, the negative and exploration surfaces, the PDF, and the no-overclaim boundary.

Result footer

```
Result ID: A13-CLASSII-SCALAR-K2K-DEGENERATE-FACE-PROJECTIVE-COMPACT-CORE-BOUNDARY
Claim:     A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Ledger ID: R-111
Status:    T4 ANALYTIC/EXACT/EXECUTED ADVANCE AND METHOD BOUNDARY;
           MIXED ALL-Q, NELSON, AND SECTOR A OPEN
Scope:     Stationary scalar circle k:2k model. All-q theorem only on vw=0;
           exact projective/floor/tail reductions for v,w>0. No full A1
```

production embedding is asserted.

Precise statement: both degenerate frequency faces obey the bare square-first inequality for every q and amplitude; the large-A mixed projective limit and first correction are positive; an exact algebraic floor removes large q and explicit tails reduce the remaining work; three separated proof shortcuts fail without refuting the target.

Dependencies: R-063, R-087, R-101, R-103--R-105, R-108--R-110.

Evidence grade: ANALYTIC/EXACT/EXECUTED; TWO NON-IMPORTING ROUTES.

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe ‘
codes/foundations/a13_classii_scalar_k2k_projective_compact_core_boundary_verify.py
```

Expected output:
primary 46/46; independent 35/35; manifest-pinned integrated and aggregate contracts; PASS.

Negative records:
NG-2026-07-28-A13-K2K-QUADRATIC-BESSEL-UPPER-DOMINATION;
NG-2026-07-28-A13-K2K-CONDITIONAL-SCALAR-TENSORIZATION;
NG-2026-07-28-A13-K2K-TILTED-VARIANCE-MONOTONICITY.

Falsification gate: failure of any scaling, face-theorem, exponential-moment, projective, positive-coefficient, floor, Bessel, tail, conditional no-go, tilted-moment, independent, hash, PDF, ledger, or scope check.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement:
the mixed v,w>0 all-q inequality, full A1 cluster, OVERLAP_src, Nelson, removals, measure, and Sector A remain open.

Next required action: prove a uniform projective remainder and certify the remaining compact mixed core by interval tails, or find a new analytic comparison; only then attempt the owner-preserving full-A1 conditional cluster and one-use source/sextic aggregation ledger.